



Solutions:

1. A. 500 m

d_1 = total dist. covered by two turtles

d_2 = dist. travelled by fly

$$t_1 = t_2$$

$$\frac{d_1}{v_{\text{net}}} = \frac{d_2}{v_{\text{fly}}}$$

$$\frac{150 \text{ m}}{10 \frac{\text{m}}{\text{s}}} + 20 \frac{\text{m}}{\text{s}} = \frac{d_2}{100 \frac{\text{m}}{\text{s}}}$$

$$d_2 = 500 \text{ m} \rightarrow \text{Ans}$$

2. D. 37.5 days

t_A = time taken for A to do work

t_B = time taken for B to do work

$$\text{rate} \propto \frac{1}{t}$$

$$\frac{\text{rate}_A}{\text{rate}_B} = \frac{t_B}{t_A} \Rightarrow \frac{6}{8} = \frac{50 \text{ days}}{t_B}$$

$$t_B = 37.5 \text{ days} \rightarrow \text{Ans}$$

3. A. 3:06.36

x = dist. traveled by min. hand

$x/12$ = dist. traveled by hour hand

Difference bet. hour and min hand

$$\text{now: } \left(\frac{x}{12} + 15 \right) - x$$

Difference bet. min and hour hand 20

$$\text{min. later: } (x + 20) - \left(\frac{x + 20}{12} + 15 \right)$$

Equating,

$$\frac{x}{12} + 15 - x = (x + 20) - \left(\frac{x + 20}{12} + 15 \right)$$

$$x = 6.36 \text{ mins.}$$

$$\therefore 3:06.36 \rightarrow \text{Ans}$$

4. D. 10

	Past	Present
Elizabeth	$18 - x$	18
Eumia	$14 - x$	14

$$(18 - x) = 2(14 - x)$$

$$18 - x = 28 - 2x$$

$$x = 10$$

5. B. 17

	Past	Present
Father	x	38
Son	0	x

$$38 - x = x - 0$$

$$x = 19$$

Two years ago, the son was $(19 - 2) = 17$

6. A. 40

70%	+	60%	=	66%
$100 - x$		x		100

$$0.70(100 - x) + 0.60(x) = 0.66(100)$$

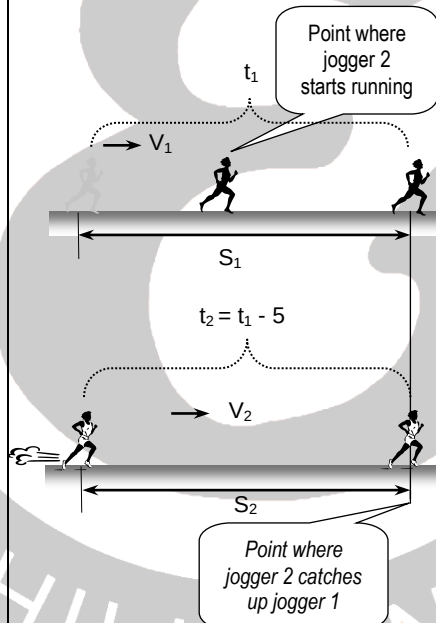
$$70 - 0.7x + 0.6x = 66$$

$$x = 40 \text{ ounces}$$

7. A. 20 min

Let: V_1 = velocity of the first jogger

V_2 = velocity of the second jogger



$$V_1 = \frac{8 \text{ km}}{\text{hr}} \times \frac{1 \text{ hr}}{60 \text{ min}} = \frac{8}{60} \text{ km/min}$$

$$V_2 = \frac{10}{60} \text{ km/min}$$

$$S_1 = S_2$$

$$V_1 t_1 = V_2 t_2$$

$$\frac{8}{60} t_1 = \frac{10}{60} (t_1 - 5)$$

$$8t_1 = 10t_1 - 50$$

$$t_1 = 25 \text{ min.}$$

$$t_2 = 25 - 5 = 20 \text{ min.}$$

8. A. 17

	Past	Present	Future
Raymond	$5(x - 6)$	?	$3(x + 5)$
Elizabeth	$x - 6$	x	$x + 5$

$$3(x + 5) - 5(x - 6) = x + 5 - (x - 6)$$

$$3x + 15 - 5x + 30 = x + 5 - x + 6$$

$$-2x + 45 = 11$$

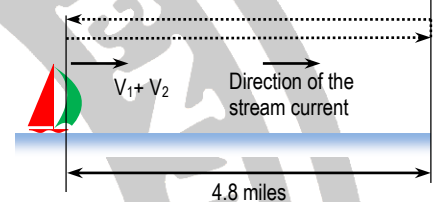
$$x = 17$$

9. C. 0.76 mile per hour

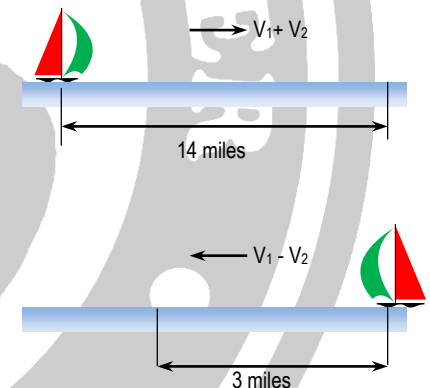
Let: V_1 = velocity of man on boat

V_2 = velocity of stream

Total time = 14 hrs



The two figures below that the same time.



$$\frac{4.8}{V_1 + V_2} + \frac{4.8}{V_1 - V_2} = 14$$

$$\frac{4.8(V_1 + V_2) + 4.8(V_1 - V_2)}{(V_1 + V_2)(V_1 - V_2)} = 14$$

$$\frac{4.8V_1 - 4.8V_2 + 4.8V_1 + 4.8V_2}{V_1^2 - V_1V_2 + V_1V_2 - V_2^2} = 14$$

$$9.6V_1 = 14V_1^2 - 14V_2^2 \rightarrow \text{Eq. 1}$$

$$\frac{14}{V_1 + V_2} = \frac{3}{V_1 - V_2}$$

$$14V_1 - 14V_2 = 3V_1 + 3V_2$$

$$11V_1 = 17V_2$$

$$V_1 = 1.545V_2 \rightarrow \text{Eq. 2}$$

Substituting equation 2 in equation 1:



$$\begin{aligned} x &= \text{no. of men} \\ y &= \text{no. of women} \\ \frac{x}{y} &= \frac{5}{3} \Rightarrow y = \frac{3}{5}x \\ \frac{x-24}{y+24} &= \frac{3}{5} \\ \frac{x-24}{\frac{3}{5}x+24} &= \frac{3}{5} \end{aligned}$$

$$\begin{aligned} x &= 60 \\ \therefore 60 - 24 &= 36 \rightarrow \text{Ans} \end{aligned}$$

21. A. 675 MW

$$\begin{aligned} P &\propto \frac{VI}{d} \\ \therefore \frac{P_1 d_1}{V_1 I_1} &= \frac{P_2 d_2}{V_2 I_2} \\ \frac{(150\text{MW})(150\text{km})}{(115\text{kV})(1000\text{A})} &= \frac{x(100\text{km})}{(230\text{kV})(1500\text{A})} \\ x &= 675\text{MW} \rightarrow \text{Ans} \end{aligned}$$

22. A. P 73.50

$$\begin{aligned} d &= 0.50; n = 12(12) = 144 \\ S &= \frac{n}{2} [2a_1 + (n-1)d] \\ 5436 &= \frac{144}{2} [2a_1 + (144-1)(0.5)] \\ a_1 &= 2 \\ a_{144} &= a_1 + 143d = 2 + 143(0.5) \\ a_{144} &= 73.50 \end{aligned}$$

23. D. 7, -7/6

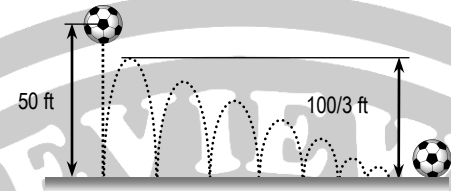
$$\begin{aligned} a_1 &= x; a_2 = 2x + 7; a_3 = 10x - 7 \\ r &= \frac{a_2}{a_1} = \frac{a_3}{a_2} \\ \frac{2x+7}{x} &= \frac{10x-7}{2x+7} \\ (2x+7)^2 &= x(10x-7) \\ 4x^2 + 28x + 49 &= 10x^2 - 7x \\ 6x^2 - 35x - 49 &= 0 \\ x &= 7, -\frac{7}{6} \end{aligned}$$

24. A. 9/2

$$a_1 = 6; a_2 = -2$$

$$\begin{aligned} r &= \frac{a_2}{a_1} = \frac{-2}{6} = -\frac{1}{3} \\ S &= \frac{a_1}{1-r} = \frac{6}{1-\left(-\frac{1}{3}\right)} = \frac{9}{2} \end{aligned}$$

25. A. 250 feet



Let: D = total distance traveled by ball

$$\begin{aligned} a_1 &= 50 \left(\frac{2}{3} \right) = \frac{100}{3} \\ S &= \frac{a_1}{r-1} = \frac{\frac{100}{3}}{1-\frac{2}{3}} = 100 \\ D &= 50 + 2S = 50 + 2(100) = 250 \text{ feet} \end{aligned}$$

26. C. 5

$$\begin{aligned} h_3 &= 15; a_3 = 1/15 \\ h_9 &= 6; a_9 = 1/6 \\ a_3 &= a_1 + 2d \\ \frac{1}{15} &= a_1 + 2d \\ a_1 &= \frac{1}{15} - 2d \rightarrow \text{Eq. 1} \\ a_9 &= a_1 + 8d \\ \frac{1}{6} &= a_1 + 8d \rightarrow \text{Eq. 2} \end{aligned}$$

Substitute equation 1 in equation 2:

$$\begin{aligned} \frac{1}{6} &= \frac{1}{15} - 2d + 8d \\ d &= \frac{1}{60} \\ \text{Substitute } d &\text{ in equation 1:} \\ a_1 &= \frac{1}{15} - 2\left(\frac{1}{60}\right) = \frac{1}{30} \\ a_{11} &= a_1 + 10d = \frac{1}{30} + 10\left(\frac{1}{60}\right) = \frac{1}{5} \end{aligned}$$

$$\text{Therefore: } h_{11} = \frac{1}{a_{11}} = \frac{1}{\frac{1}{5}} = 5$$

27. A. 58 centavos

$$a_1 = 40; a_5 = 76$$

$$\begin{aligned} a_5 &= a_1 + 4d \\ 76 &= 40 + 4d \\ d &= 9 \\ \text{Therefore:} \\ a_2 &= 40 + 9 = 49 \\ a_3 &= 49 + 9 = 58 \\ a_4 &= 58 + 9 = 67 \\ \text{Average} &= \frac{40 + 49 + 58 + 67 + 76}{5} = 58 \end{aligned}$$

28. C. 0.9122

Let: x = total volume of air in the tank
y = total volume removed from the tank after the 6th stroke

$$\text{Vol. left after 1st stroke} = x - \frac{1}{3}x = \frac{2}{3}x$$

$$\begin{aligned} \text{Vol. left after 2nd stroke} \\ &= \frac{2}{3}x - \frac{1}{3}\left(\frac{2}{3}x\right) = \frac{4}{9}x \end{aligned}$$

$$\begin{aligned} \text{Vol. left after 3rd stroke} \\ &= \frac{4}{9}x - \frac{1}{3}\left(\frac{4}{9}x\right) = \frac{8}{27}x \end{aligned}$$

Note: By inspection, the volume left after each stroke forms a GP whose common

$$\text{ratio is } r = \frac{\frac{4}{9}}{\frac{2}{3}} = \frac{2}{3}$$

Solving for the volume left after the 6th stroke

$$a_6 = a_1 r^5 = \left(\frac{2}{3}x\right)\left(\frac{2}{3}\right)^5 = 0.08779x$$

Thus, the total volume removed after the 6th stroke: y = x - 0.08779x = 0.9122x

29. A. 9.87 hours

work done = work load

$$\begin{aligned} \frac{\text{no. of workers} \times \text{time of doing the job}}{\text{no. of job done}} \\ &= \frac{\text{no. of workers} \times \text{time to finish the job}}{\text{no. of job to do}} \end{aligned}$$

$$\begin{aligned} \left[\frac{8 \text{ men} \times 7 \text{ hrs}}{150 \text{ ft}} + \frac{3 \text{ men} \times 4 \text{ hrs}}{100 \text{ ft}} \right] &= \frac{10 \text{ men} \times 'x' \text{ hrs}}{200 \text{ ft}} \\ x &= 9.87 \text{ hrs} \end{aligned}$$

30. A. 334



Rate of work of the three men:

$$\text{Atan} = \frac{1}{15}$$

$$\text{Marlon} = \left(\frac{1}{15}\right)0.50 + \frac{1}{15} = \frac{1}{10}$$

$$\text{Danny} = 2\left(\frac{1}{15}\right) = \frac{2}{15}$$

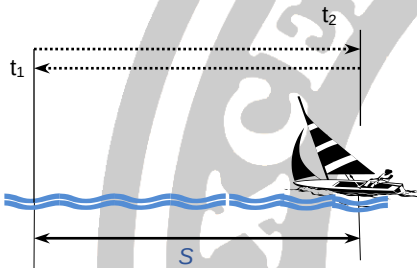
$$\text{First half: } \frac{1}{15}(1.5 + t_1) + \frac{1}{10}t_2 = \frac{1}{2} \Rightarrow t_1 = 2.4$$

$$\text{Second half: } \left(\frac{1}{15} + \frac{1}{10} + \frac{2}{15}\right)t_2 = \frac{1}{2} \Rightarrow t_2 = \frac{5}{3}$$

Solving for Atan's total time of working, t:

$$t = 1.5 + 2.4 + \frac{5}{3} = 5.5667 \text{ hr} = 334 \text{ min}$$

31. A. 2.5 miles



$$\text{time} = \frac{\text{distance}}{\text{velocity}}$$

$$t_1 + t_2 = t_{\text{total}}$$

$$\frac{S}{5} + \frac{S}{2} = \frac{7}{4}$$

$$S = 2.5 \text{ miles}$$

32. C. 47

Let: t = ten's digit of the number

u = unit's digit of the number

$$\text{Number} = 10t + u$$

$$t = u - 3 \rightarrow \text{Eq. 1}$$

$$\frac{10t + u}{t + u} = 4 + \frac{3}{t + u}$$

$$\frac{10t + u - 3}{t + u} = 4$$

$$10t + u - 3 = 4t + 4u$$

$$6t - 3 = 3u \rightarrow \text{Eq. 2}$$

Substituting equation 1 in equation 2:

$$6(u - 3) - 3 = 3u$$

$$6u - 18 - 3 = 3u$$

$$3u = 21$$

$$u = 7$$

$$t = 7 - 3 = 4$$

$$\text{Number} = 10(4) + 7 = 47$$

33. C. 36/55

Let: $\frac{x}{y}$ = the fraction

$$\frac{x-3}{y} = \frac{3}{5}$$

$$3y = 5x - 15$$

$$y = \frac{5}{3}x - 5 \rightarrow \text{Eq. 1}$$

$$\frac{x}{y-1} = \frac{2}{3}$$

$$3x = 2y - 2 \rightarrow \text{Eq. 2}$$

Substituting equation 1 in equation 2:

$$3x = 2\left[\frac{5}{3}x - 5\right] - 2$$

$$3x = \frac{10}{3}x - 10 - 2$$

$$0.333x = 12$$

$$x = 36$$

$$y = \frac{5}{3}(36) - 5 = 55$$

$$\text{Thus, the fraction is } \frac{36}{55}$$

34. B. 6 kg

$$\begin{array}{|c|} \hline 4\% \\ \hline 10 \end{array} - \begin{array}{|c|} \hline 0\% \\ \hline x \end{array} = \begin{array}{|c|} \hline 10\% \\ \hline 10-x \end{array}$$

$$0.04(10) - 0(x) = 0.10(10 - x)$$

$$0.4 = 1 - 0.1x$$

$$x = 6 \text{ kg}$$

35. A. 19

Let: x = number of days needed by A, B and C to finished the work working together.

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = \frac{1}{x}$$

$$\frac{1}{A} + \frac{1}{B} = \frac{1}{42} \rightarrow \text{Eq. 1}$$

$$\frac{1}{B} + \frac{1}{C} = \frac{1}{31} \rightarrow \text{Eq. 2}$$

$$\frac{1}{A} + \frac{1}{C} = \frac{1}{20} \rightarrow \text{Eq. 3}$$

Adding the three equations:

$$\left[\frac{1}{A} + \frac{1}{B}\right] + \left[\frac{1}{B} + \frac{1}{C}\right] + \left[\frac{1}{A} + \frac{1}{C}\right] = \frac{1}{42} + \frac{1}{31} + \frac{1}{20} = 0.106$$

$$\frac{2}{A} + \frac{2}{B} + \frac{2}{C} = 0.106$$

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} = 0.053 = \frac{1}{x}$$

$$x = 18.87 \text{ days}$$

$$x \approx 19 \text{ days}$$

36. C. 18 hours

Let: x = time for Jimmy to finish the job

y = time for Sonny to finish the job

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{6} \rightarrow \text{Eq. 1}$$

$$\frac{1}{x} = 2\left(\frac{1}{y}\right) \rightarrow \text{Eq. 2}$$

Substituting equation 2 in equation 1:

$$\frac{2}{y} + \frac{1}{y} = \frac{1}{6}$$

$$y = 18 \text{ hours}$$

37. B. $\frac{xy}{x+y}$

Let: n = number of days needed to complete the work

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$$

$$\frac{1}{n} = \frac{x+y}{xy} \Rightarrow n = \frac{xy}{x+y}$$

38. A. 7 hours

Let: x = time needed to complete the work

$$\frac{1}{20} + \frac{1}{11} = \frac{1}{x}$$

$$x = 7.096 \text{ hours}$$

39. B. 6

Let: x = age of the girl

y = age of her brother

z = age of her sister



$$\begin{aligned}y &= 3x && \rightarrow \text{Eq. 1} \\z &= x + 8 && \rightarrow \text{Eq. 2} \\x + y + z &= 38 && \rightarrow \text{Eq. 3}\end{aligned}$$

Substituting equations 1 & 2 in equation 3:

$$\begin{aligned}x + 3x + (x + 8) &= 38 \\x &= 6\end{aligned}$$

40. A. 114

34, a_2 , a_3 , a_4 , 42

$$a_5 = a_1 + 4d$$

$$42 = 34 + 4d$$

$$d = 2$$

Thus, $a_2 = 36$, $a_3 = 38$ and $a_4 = 40$

$$\text{Sum} = 36 + 38 + 40$$

$$\text{Sum} = 114$$

41. B. 675

$$P = k \frac{VI}{d}$$

When $P = 150$, $V = 115$, $I = 100$ & $d = 150$

$$150 = k \frac{(115)(100)}{150}$$

$$k = 1.956$$

When $V = 230$, $I = 150$ and $d = 100$

$$P = 1.956 \left[\frac{(230)(150)}{100} \right] = 675 \text{ megawatts}$$

42. B. 16

$$x = k \frac{y}{z}$$

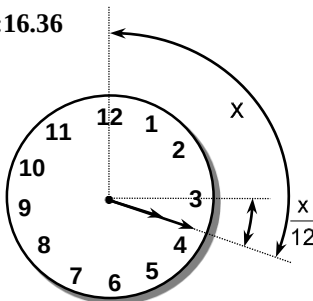
When $x = 14$, $y = 7$ and $z = 2$

$$14 = k \frac{7}{2}$$

$$k = 4$$

When $y = 16$ and $z = 4$, $x = (4) \left(\frac{16}{4} \right) = 16$

43. D. 3:16.36



$$x = 15 + \frac{x}{12} = 16.36 \text{ min}$$

Therefore, the time is 3:16.36

44. D. 7 hours and 50 minutes

$$\frac{1}{9} + \frac{1}{12} - \frac{1}{15} = \frac{1}{x}$$

$$x = 7.826 \text{ hours}$$

$$x = 7 \text{ hours \& 50 minutes}$$

45. B. 63

x = no. of eggs at start

$$\text{First time: } x - \left(\frac{x}{2} + \frac{1}{2} \right) = \frac{x-1}{2}$$

$$\text{Second time: } \frac{\left(\frac{x-1}{2} \right) - 1}{2}$$

$$\text{Third time: } \frac{\left(\frac{\left(\frac{x-1}{2} - 1 \right)}{2} \right) - 1}{2} = 7$$

Using SHIFT $\rightarrow$ CALC

$$x = 63 \rightarrow \text{Ans}$$

46. C. 1.2 hr

d_1 = distance traveled by Cathy

d_2 = distance traveled by Nina

t = time taken for both to travel

$$d_1 + d_2 = 24$$

$$v_1 t + v_2 t = 24$$

$$(8\text{mph} + 12\text{mph}) t = 24$$

$$t = 1.2 \text{ hrs.} \rightarrow \text{Ans}$$

47. B. 5

p = no. of pennies

d = no. of dimes

q = no. of quarters

$$p + d + q = 44 \rightarrow \text{eq.1}$$

$$0.01p + 0.10d + 0.25q = 5.9 \rightarrow \text{eq.2}$$

$$d = 2q$$

$$\therefore d - 2q = 0 \rightarrow \text{eq.3}$$

Using MODE 5 2,

$$p = 5 \rightarrow \text{Ans}$$

$$d = 26$$

$$q = 13$$

48. B. 784 feet

$$a_1 = 4; d = 8; n = 14$$

$$S = \frac{n}{2} [2a_1 + (n-1)d]$$

$$S = \frac{14}{2} [2(4) + 13(8)] = 784$$

49. B. 60

$$a_1 = 30; a_5 = 90$$

$$a_5 = a_1 + 4d$$

$$90 = 30 + 4d$$

$$d = 15$$

$$S = \frac{n}{2} [2a_1 + (n-1)d]$$

$$S = \frac{5}{2} [2(30) + 4(15)] = 300$$

$$\text{Average score} = \frac{300}{5} = 60$$

50. D. 1/2

$$a_1 = 1/3; a_2 = 1/9; a_3 = 1/27$$

By inspection, $r = 1/3$

$$S = \frac{a_1}{1-r} = \frac{\frac{1}{3}}{1-\frac{1}{3}} = \frac{1}{2}$$